

Wellington Santos

Machine Learning Engineer

santossowellington23@gmail.com · +55 19983546558 · Gonçalves Dias, Maranhão, Brazil · www.linkedin.com/in/welington-santos-53044a3b2/

PROFESSIONAL SUMMARY

Machine Learning Engineer with **8+** years across fintech, e-commerce, and healthcare, specializing in production ML systems and edge inference. Proven expertise in building anomaly detection and classification models using **PyTorch** and **TensorFlow**, deploying high-throughput services with sub-100ms latency, and orchestrating data pipelines with **Airflow** and **Spark**. Ready to weaponize AI for real-time defense mechanisms protecting billions of transactions.

SKILLS

Languages: Python, SQL, TypeScript, Rust, JavaScript, HTML, CSS
Generative AI: LangChain, LlamaIndex, OpenAI API, Anthropic API
Data & ML: PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, Spark, Airflow, dbt, MLflow
Backend: FastAPI, Node.js, Express, Django, gRPC
Database: PostgreSQL, MongoDB, Redis, Elasticsearch, DynamoDB
Testing: Pytest, Jest, Playwright, k6
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions, CloudFront, Lambda, CloudWatch, EC2, S3
Others: RESTful API, GraphQL, WebSocket, Git, Jira, Confluence, Grafana, Prometheus, Datadog, Agile

EXPERIENCE

Fanatics Oct 2023–Present
Senior ML Engineer

- Led the development of a real-time recommendation engine using **PyTorch** and **Spark**, processing 500k+ daily events to personalize sports collectibles and betting recommendations across the fan ecosystem, serving traffic spikes during major sporting events.
- Spearheaded the design and deployment of a fraud detection model using **TensorFlow** and **Scikit-Learn** that identified suspicious betting patterns and account takeover attempts, reducing fraudulent transactions by **32%** while maintaining a false-positive rate below 2%.
- Owned end-to-end ML infrastructure improvements: refactored data ingestion pipelines using **Airflow** and **Kafka**, cutting feature-generation latency from 6 hours to 45 minutes and enabling daily model retraining with confidence.
- Implemented automated model monitoring with **Prometheus** and **Grafana**, tracking data drift and performance degradation across three production models, which caught a 15% accuracy drop before it impacted business metrics.

- Collaborated with security engineers to build an anomaly detection layer using **LangChain** agents that flagged unusual user behavior patterns in real-time, integrating with the account verification pipeline to reduce account compromise incidents by **28%**.
- Optimized inference latency for a classification model deployed on **AWS Lambda** by quantizing weights and caching embeddings in **Redis**, reducing p99 prediction time from 280ms to **67ms** across 2.3M daily predictions.
- Trained a multi-class classifier using **PyTorch** on historical attack simulation data provided by the offensive security team, achieving **94%** precision on zero-day betting bot detection.
- Mentored two junior ML engineers on production deployment best practices, pair-programming on **Docker** containerization and **Kubernetes** orchestration of inference services, which accelerated their independent project delivery by three weeks.
- Documented attack vector taxonomies and threat models in **Confluence**, creating a shared reference for both security and ML teams to align on training data labeling and model evaluation metrics.
- Evaluated **Cloudflare Workers** and AWS edge computing for deploying lightweight anomaly detection models closer to traffic sources, completing a proof-of-concept that demonstrated **12%** latency improvement over centralized inference.
- Shipped a batch prediction job that scored 8M+ historical user sessions nightly using **Spark**, generating features for the next day's model training and populating dashboards used by the fraud team for case investigation.

Walmart

Sep 2022–Oct 2023

Senior Backend Engineer

- Drove the implementation of a demand forecasting pipeline using **TensorFlow** and **PostgreSQL**, enabling inventory teams across 4,500+ stores to reduce stockouts by **18%** while cutting excess inventory by **12%** during peak seasons.
- Architected a scalable feature store using **Spark** and **dbt**, centralizing ML-ready datasets for pricing optimization and customer segmentation models serving 150M+ weekly predictions across Walmart.com and Sam's Club platforms.
- Built a real-time price elasticity model using **PyTorch** that ingested click-stream data via **Kafka**, enabling dynamic pricing adjustments that improved revenue per transaction by **7.2%** during a three-month A/B test.
- Designed and deployed a model monitoring framework using **MLflow** and **Grafana**, tracking performance metrics for eight production models across retail, e-commerce, and advertising domains, reducing model staleness incidents from monthly to under **3** per quarter.
- Optimized a customer churn prediction model inference pipeline using **FastAPI** and **Redis** caching, reducing API response time from 420ms to **85ms**, which enabled real-time personalized retention offers in the mobile app.
- Collaborated with the data governance team to build an automated data quality pipeline in **Airflow**, validating training datasets for concept drift and missing values before model retraining, preventing **five** instances of degraded model performance.

- Evaluated **LlamaIndex** and **LangChain** for building an intelligent customer support chatbot trained on historical FAQ data, delivering a working prototype that reduced support ticket volume by **14%** in a pilot.
- Mentored four data engineers on **SQL** query optimization and **Spark** best practices, improving end-to-end data pipeline execution time by **35%** and establishing baseline performance standards for the team.

E-POT

Jun 2018–July 2022

Full Stack Engineer

- Built a full-stack patient risk stratification system for healthcare providers using **Python**, **FastAPI**, and **PostgreSQL**, integrating **PyTorch** models that predicted readmission risk with **89%** sensitivity, preventing ~2,000 avoidable hospitalizations annually.
- Shipped a real-time anomaly detection service for medical device IoT telemetry using **TensorFlow** and **Kafka**, processing 50k+ sensor readings per second and alerting clinicians to equipment failures before patient impact, reducing mean time to diagnosis by **6 hours**.
- Designed and implemented a data pipeline using **Spark** and **Airflow** that ingested EHR data, claims, and lab results into a unified warehouse, enabling downstream ML models for fraud detection in billing systems.
- Developed a financial forecasting model using **Scikit-Learn** and **NumPy** for healthcare finance operations, reducing budget variance by **22%** and enabling more accurate resource allocation across 50+ clinics.
- Integrated **MLflow** for experiment tracking and model versioning, establishing reproducibility standards across a team of five data scientists and reducing time-to-production for new models from 8 weeks to 3 weeks.
- Optimized inference performance for a critical disease classification model deployed on-premise at partner hospitals using **Docker** and **Kubernetes**, reducing prediction latency from 550ms to **180ms**, enabling real-time screening workflows.
- Maintained and improved an automated retraining pipeline using **Airflow** that monitored model performance on new patient data weekly, triggered retraining when accuracy dropped below thresholds, and validated model quality before production deployment.
- Debugged a production incident where a model began making systematically biased predictions — investigated data drift using **Pandas** and **NumPy**, identified a labeling artifact in the recent training set, and retrained the model with stratified sampling to restore fairness metrics to acceptable ranges.

EDUCATION

University of Tokyo

Apr 2014–Mar 2018

Bachelor of Science in Computer Science